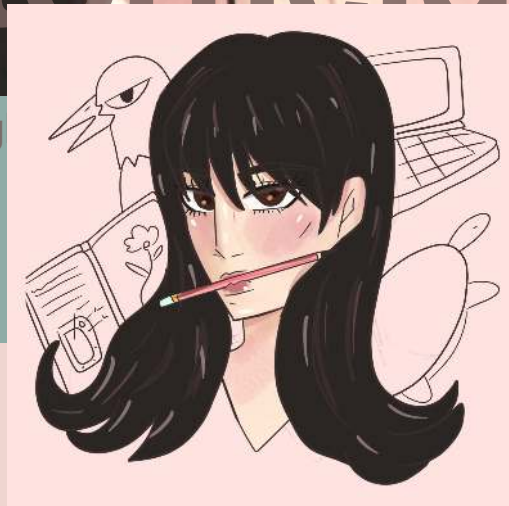


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UNIT 1 & 2

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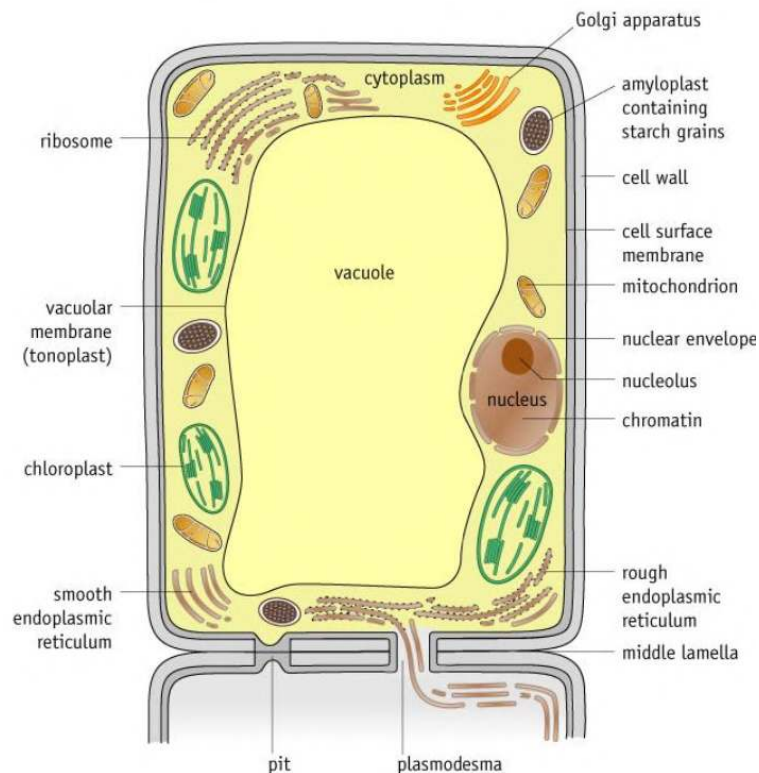
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Plants

The Plant Cell

▼ What is the basic structure of a plant cell?



1. Cell Wall: Rigid, **cellulose** outer layer of the plant cell that gives the cell its **shape** and the plant **strength** and **support**. It is **fully permeable to water-soluble substances** and **contains pores that allow the movement of substances** in and out of the cell.
2. Permanent Vacuole:
 - Membranous sac that contains Cell sap - a concentrated solution of salts, sugars, pigment, which is important in determining osmotic properties of the cell.
 - **Cell sap allows plant cells to maintain its turgidity** and stand upright by allowing water to move into the cell via osmosis by

keeping the cytoplasm pressed against the cell wall.

- **Stores** substances such as **pigments** (betacyanin in beetroot), **proteins**, lytic **enzymes**, etc. and **isolates unwanted chemicals** from the rest of the cell.

3. Chloroplasts:

- Large flattened, double-membraned structure, with a biconcave shape. The inner membrane is folded and stacked into thylakoids to increase the surface area of enzyme-controlled chemical reactions.
- Diameter 4-10 um, 2-3 um thick
- **Contains DNA**
- **Contains chlorophyll** - a green pigment that absorbs light for photosynthesis.
- **Site of Photosynthesis where solar energy is converted into chemical energy.**

4. Amyloplast:

- **Storage vacuoles** that contain starch grains.
- They **synthesise and store starch molecules**, and break them down to glucose when the cell requires it for energy.

5. Tonoplast:

- Specialised **membrane surrounding the permanent vacuole** that **controls the movement of substances in and out of the cell sap**.

6. Plasmodesmata:

- **Cytoplasmic bridges/channels between adjacent cell walls** that allow **communication** and **exchange of substances** between cells

7. Pits:

- **Sections of the cell wall with a thin layer of cellulose** where plasmodesmata are found.
- Allows the exchange of fluid and communication between adjacent cells.
- In xylem cells they allow the exchange of water

8. Middle Lamella:

- Layer of pectin that **lies between adjacent cells**
- **Cements adjacent plant cells together** by sticking the cellulose of adjacent primary cell walls together.
- Contains pectins e.g. calcium and magnesium pectates.

9. Cytoplasm:

- Internal fluid component of the cell containing essential materials i.e. proteins, water, etc.
- Site of metabolic reactions

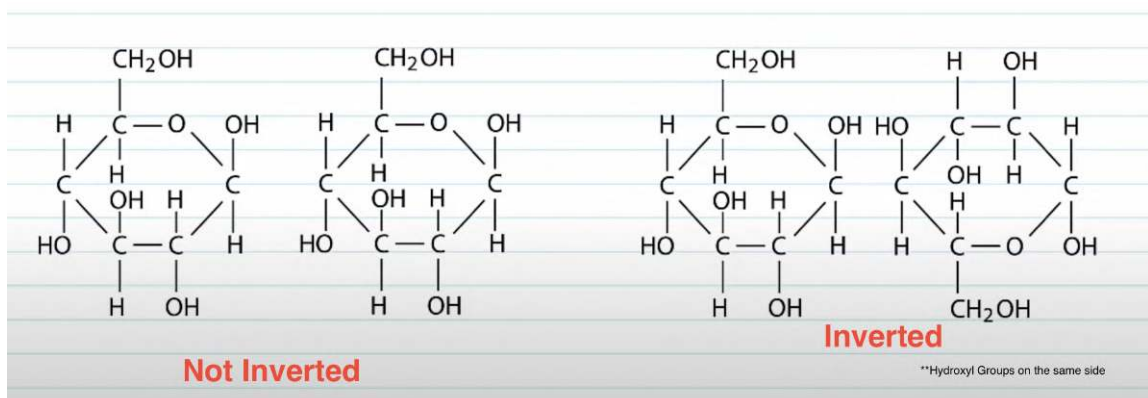
▼ How are cellulose and microfibrils formed?

What is Cellulose?

A **beta glucose polysaccharide**, where the the beta glucose monomers are held together by **1,4 glycosidic bonds** formed in a series of **condensation reactions** with the removal of water.

How is Cellulose Formed?

- Series of condensation reactions of beta glucose monomers
- Forms 1,4 glycosidic bonds only (no side branching)
- To form the 1,4 glycosidic bonds, each of the beta glucose molecules are inverted 180 degrees from the previous molecule.
- This prevents coiling and forms unbranched, long, straight chains of cellulose



- The straight chains of cellulose run parallel to each other such that the OH groups of adjacent cellulose molecules to be in close proximity of

each other.

- This allows hydrogen bonds to form between OH groups on adjacent chains. This results in **crosslinking** between cellulose chains, which makes the cellulose collectively strong.

Formation of Cellulose Microfibrils:

Cellulose molecule held together by hydrogen bonds form microfibrils.

Microfibrils are **arranged parallel** to each other to form a **mesh-like structure**. In the cell wall, these microfibrils are held in a **matrix** of different substances.

***Alpha Glucose - Hydroxyl Group on the Bottom of the Ring on carbon 1,
Beta Glucose → top***

Starch VS Cellulose

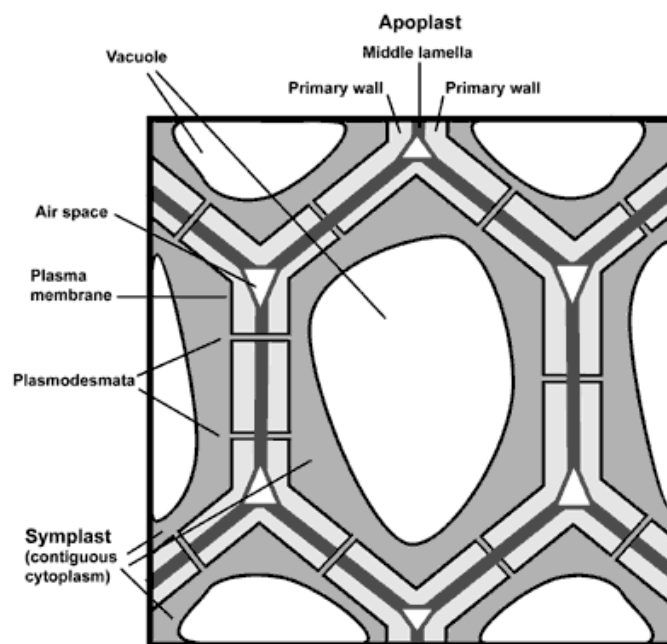
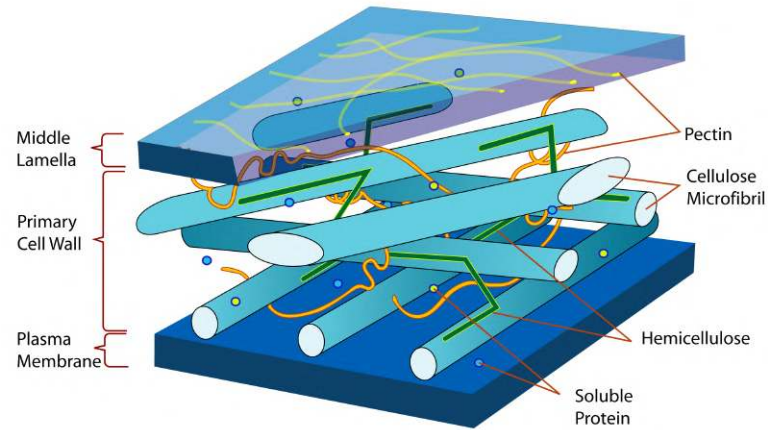
Starch	Cellulose
Made up of glucose monomer.	Made up of β glucose monomers.
Contains 1,4 and 1,6 glycosidic bonds i.e. there is side-branching.	Contains 1,4 glycosidic bonds only i.e. no side-branching.
Used as a storage carbohydrate.	Used as a structural carbohydrate to form the cell wall.
Winds into a spiral shape.	Remains as a long, straight chain.

Cellulose as a Fibre:

- Cellulose is the main component of plant walls, making it the most abundant organic material in the world.
- Cellulose is not an easily digestible food source because it is very hard to break down by hydrolysis. This is because most animals lack the cellulase enzyme needed to break the 1,4 glycosidic bonds between β -glucose molecules.
- Some herbivores have symbiotic bacteria in their guts that produce the enzyme. This allows herbivores to digest a larger proportion of their diet and gain more energy.

- Cellulose is still important in the human diet as it provides the fibre needed to keep the digestive system healthy.

▼ Explain the structure of the plant cell wall



1. Middle Lamella:

- The first layer that is formed when plant cells divide (the cell plate forms the middle lamella)
- **Made up of Pectin** (polysaccharide), which acts like a glue that **holds/binds adjacent plant cells together**.
- Pectin contains negatively charged carboxyl ($-\text{COOH}$) groups which combine with positive calcium ions to form calcium pectate.

Calcium pectate binds the cellulose of the primary walls on either side.

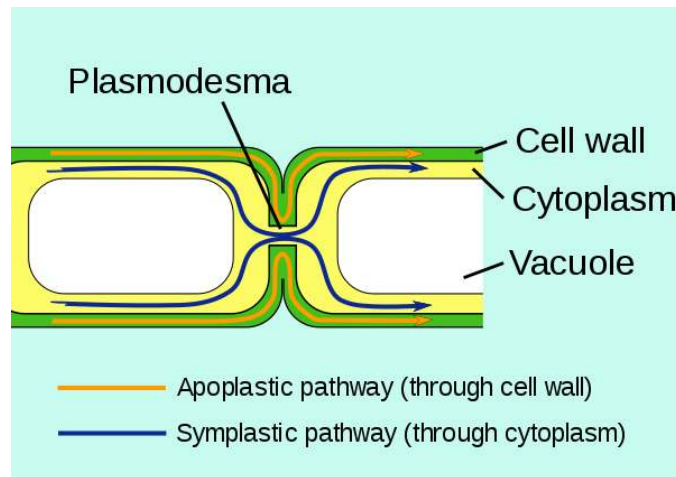
2. Primary Cell Wall:

- Composed of **cellulose microfibrils (cellulose molecules held together by hydrogen bonds)** that build up on either sides of the **middle lamella**.
- Microfibrils are **arranged in a similar direction** (gives flexibility), **parallel** to each other in a **crisscrossed pattern**.
- **Microfibrils held within a matrix of proteins, pectin and hemicellulose**. Forms a composite material that is turgid → gives cell strength and support.
- Provides the strength and flexibility needed to allow for cell growth.

3. Secondary Cell Wall:

- Only **formed between the primary cell wall and plasma membrane** in some plant cells.
- Forms when the primary cell wall has stopped dividing and growing.
- Formed by a **buildup of cellulose microfibrils laid densely at different angles** (Makes the cell wall more compact and rigid)
- Hemicellulose added to harden the cell wall. Lignin added to woody plants, suberin added to cork → both make cell wall impermeable.
- **Lignin holds the microfibrils parallel to each other in a fixed arrangement.**
- Cells with heavily lignified cell walls - fibres

▼ Describe the plasmodesmata



The plasmodesmata are **cytoplasmic bridges** between adjacent cells that **allow the communication and exchange of materials between cells**.

They are formed as the cell divides; the two cells do not separate completely and threads of cytoplasm remain between them. The threads pass through the gaps of the newly formed cell walls and signalling substances can pass from one cell to another through the cytoplasm. The cell walls are thinner at the region of the plasmodesmata.

Symplast - Interconnected Cytoplasm

When secondary thickening occurs, hemicellulose and lignin are deposited in the cell wall making it thicker. In areas around plasmodesmata, this thickening does not occur, leaving thin areas of the cytoplasm known as **pits**. In xylem cells where there is no cytoplasm, the pits allow water to move between the xylem vessels.

Plant Cells & Tissues

▼ What are the functions of the stem?

1. To **provide support** to the plant
2. To Elongate and orient the shoot in a way that holds leaves in a favourable position to receive light for photosynthesis.
3. Elevates reproductive structures/holds the flower in a favourable position for pollination
4. Movement of materials around the plants: Stems provide the route through which the products of photosynthesis are carried from the leaves (site of photosynthesis) to the rest of the plant where it is needed.

5. Photosynthesis: Green Stems contain chlorophyll and carry out little photosynthesis.

▼ Differentiate between the 3 Specialised Plant Cells

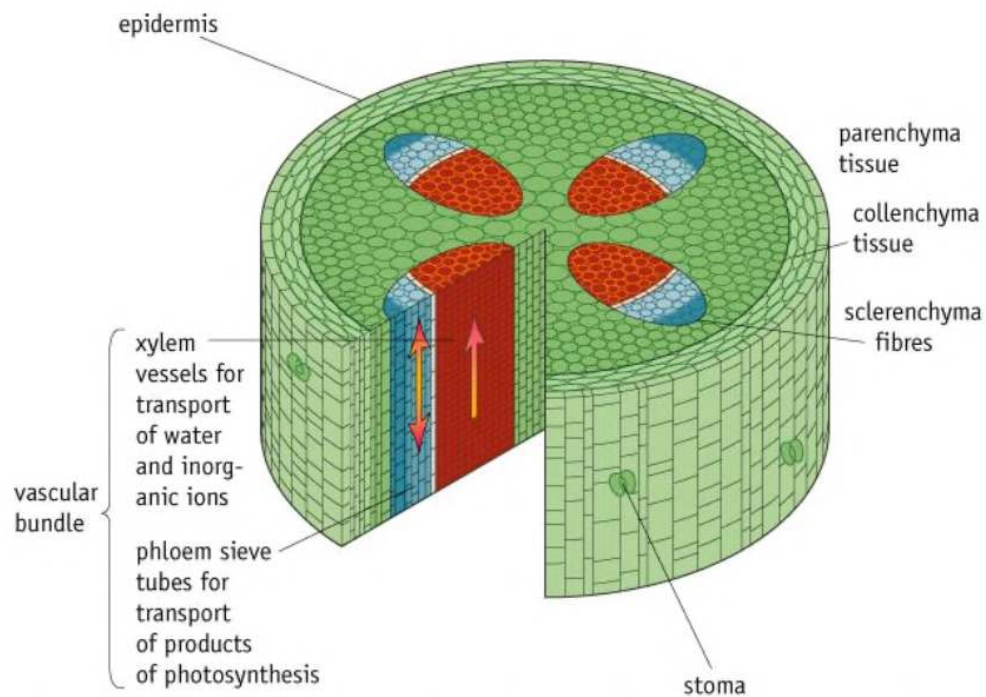
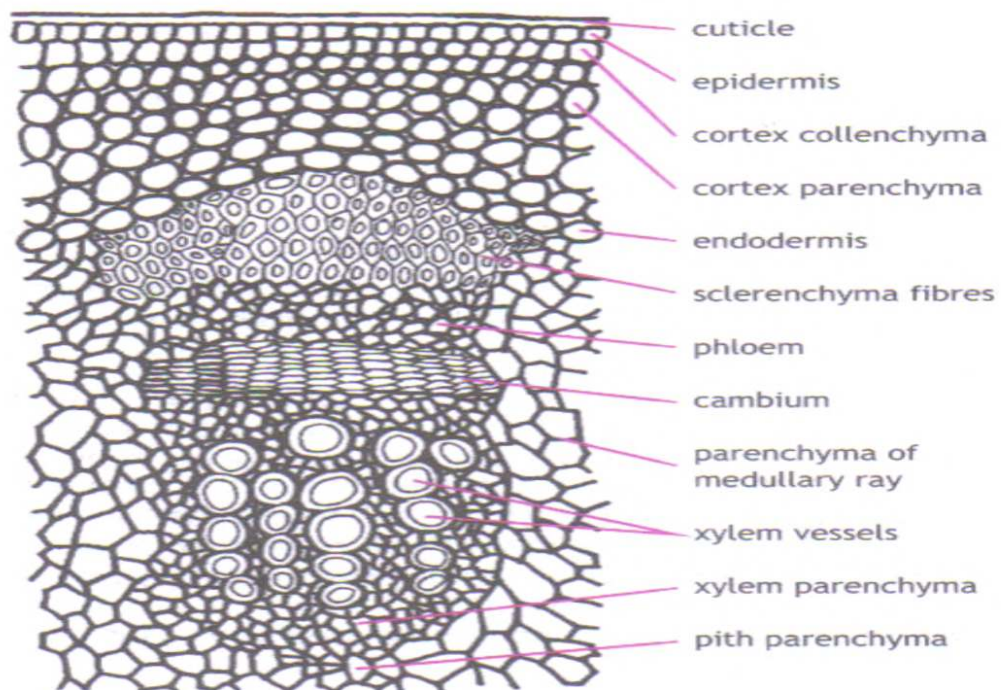


Figure 4.13 – The general arrangement of vascular tissue within the stem of a dicotyledon. The arrows represent the direction of flow through the xylem and phloem.

Transverse section of a young dicot stem in detail



Differentiated Cells

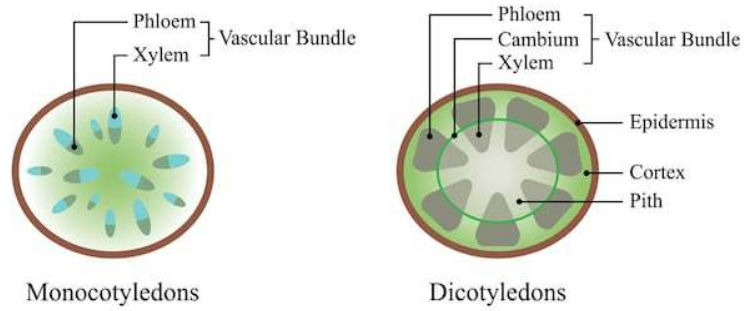
Property	<u>Aa</u> Parenchyma	<u>Ee</u> Collenchyma	<u>Ee</u> Sclerenchyma
Location	<u>Found in the soft parts of the plant</u>	Found in the leaves, stems, and petioles	Found in the mature parts of plants/trees
Type of Cell	<u>Unspecialised, Living cells</u>	Specialised, living Cells	Specialised, non-living Cells
Cell Wall	<u>Thin cellulose cell wall</u>	Uneven cell wall made of cellulose, pectin and hemicellulose	Rigid, thick cell wall made up of waterproofing lignin
Secondary Cell Wall	<u>No secondary cell wall</u>	Some secondary thickening	Strong, lignified secondary walls made of cellulose microfibrils positioned at right angles to each other.

Property	<u>Aa</u> Parenchyma	Collenchyma	Sclerenchyma
Intercellular Spaces between cells	<u>Intercellular Space present between cells</u>	Little to Intercellular Space present between cells	No Intercellular Space present between cells
Function	<u>Act as packing in stems and roots to give support. Help in storage of foods, in gaseous exchange, and in photosynthesis.</u>	Provide mechanical support to the plant. Resists bending and stretching by the wind. Gives flexible support without restraining growth.	Provides mechanical support to the plant. It also supports transportation of water and nutrients to the plants. Develops as the plant grows to support the increasing weigh of the upper part of the plant
Maturity	<u>Untitled</u>	At maturity, cells remain living and flexible; they elongate with the stems and leaves they support	Dies at maturity due to high lignification — water can no longer pass through the lignin

▼ Explain how Sclerenchyma Cells make Fibre and Sclerids

- Some Sclerenchyma cells make fibres - long cells found in bundles positioned outside of a stem or root. Lignin is deposited on the cell walls of these fibres in a spiral/ring pattern, making the fibres strong and flexible. When the fibre is lignified, the cell contents die because water cannot pass through the lignin, so the fibres become hollow tubes.
- Sclerenchyma cells can become impregnated with lignin to form sclereids. Sclerids are more boxier and irregular in shape than fibre. These tough cells may be found in groups throughout the cortex of the stem or individually in plant tissues.

▼ Explain the structures and functions of the Vascular Bundle



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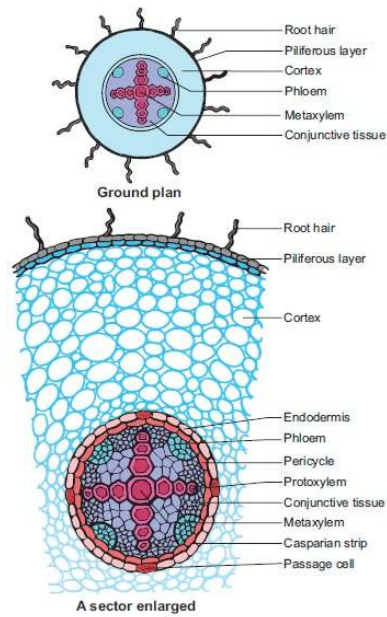
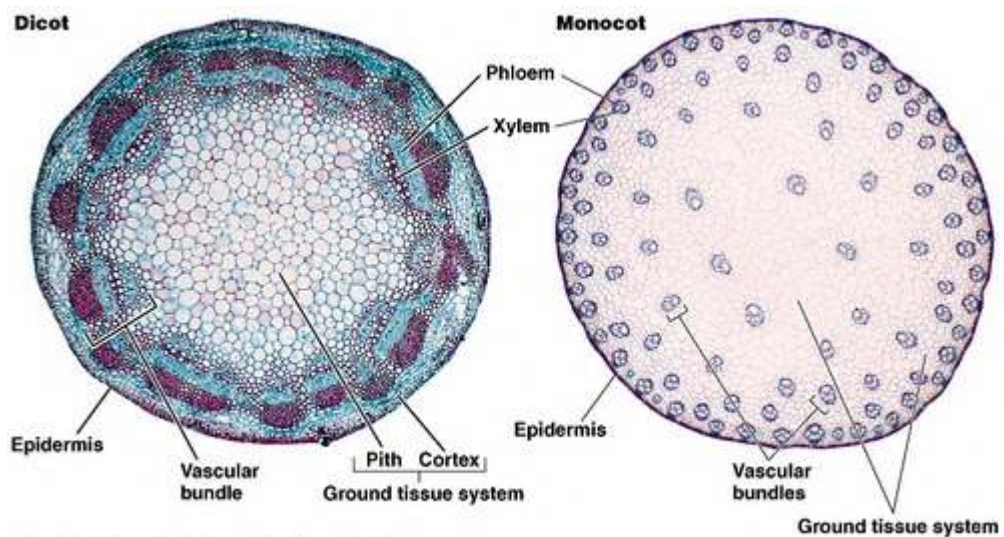
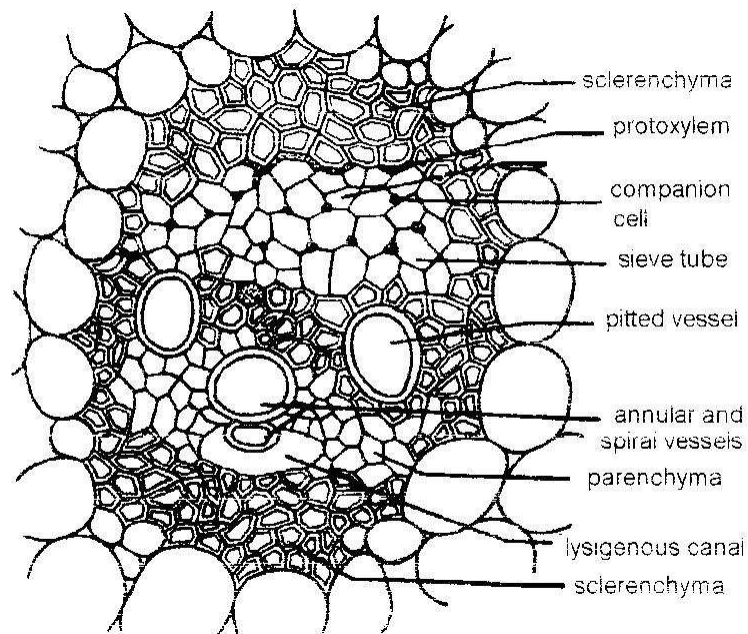


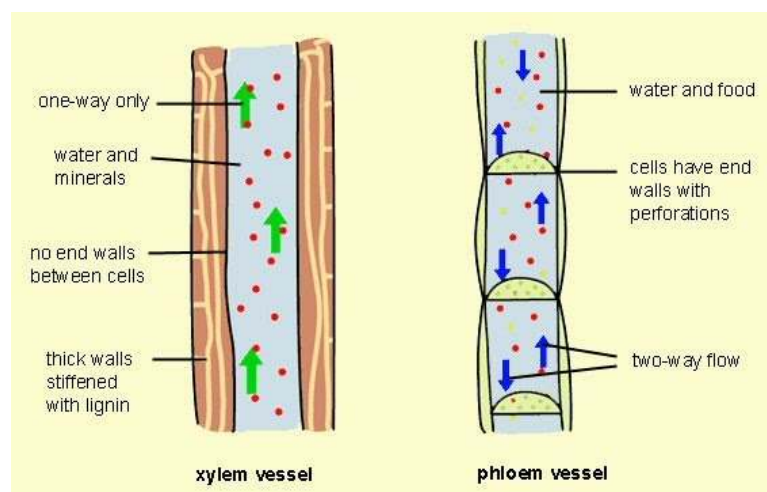
Figure 9.19: T.S. of Dicot root (Bean root)



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- Monocotyledons - Vascular bundle is scattered
- Dicotyledon - The vascular bundle forms a ring shape around the stem, has cambium



Vascular Bundle

Aa Property	≡ Xylem	≡ Phloem	≡ Sclerenchyma
----------------	---------	----------	----------------

Aa Property	☰ Xylem	☰ Phloem	☰ Sclerenchyma
<u>Function</u>	Transports water and dissolved mineral ions from the roots to the photosynthetic parts of the plant. Also supports the plant	Transports organic solutes (sucrose) from the leaves to where it is needed for growth or storage as starch (Translocation)	Provides support and tensile strength
<u>Description</u>	Bundles of hollow, longitudinal, dead cells	Living cells joined end to end forming a sieve tube element.	Bundles of dead cells that run vertically up the stem
<u>End walls</u>	Absent	Present	Present
<u>Cell walls</u>	Stiffened/reinforced with Lignin in various patterns (spiral, circular, scalariform and pitted)	No lignin	Reinforced with lignin and cellulose
<u>Direction of Flow</u>	One Way, Upwards	Two-Way	
<u>Type of Cells</u>	Dead	Living	Dead
<u>Features</u>	Contain tubular structures called xylem vessels which are its main functional units	Sieve tube elements with sieve plates: end walls with holes (perforated). Function: Allows Phloem sap (sugar solution) to pass through in mass flow	No Pits, Hollow lumen with end walls
<u>Untitled</u>	Has pits; allows water and mineral ions to move in and out of the vessel)	Thin layer of cytoplasm, lacks nucleus and other organelles	
<u>Untitled</u>	Hollow lumen; no cytoplasm or end walls	Connected with adjacent companion cells by plasmodesmata. Companion cells provide sieve cells with essential substances such as ATP	

Xylem Vessels:

Formation

- Starts off as protoxylem, which can stretch and grow as the walls are not fully lignified
- Cellulose microfibrils arranged vertically to increase the strength of the tube and resist compression forces from the weight of the plant
- Lignin is incorporated more and more as the stem ages, and the cells stop growing.
- Cells become impermeable to water and other substances. Cell contents die.
- Lignified xylem tissue = metaxylem
- End Walls break down, forming hollow tubes.

2.4.5 – COMPARISON OF XYLEM VESSELS AND SCLERENCHYMA FIBRES

Xylem vessels	Sclerenchyma fibres
• Made up of large cells with thick cell walls • Form a column of cells to transport water and inorganic ions • Waterproofed and strengthened by the polymer lignin laid down in spirals or rings • Dead tissue formed from previously living cells	• Elongated cells • Sole function to provide support and mechanical strength • Cell wall heavily thickened with lignin which provides great tensile and compressional strength* • Dead tissue formed from previously living cells
*tensile strength means it doesn't break easily on stretching; compressional strength means it doesn't buckle easily.	

Phloem Vessels:

Function: Translocation - transport of the dissolved products of photosynthesis.

Formation

- Living cells are lined end to end forming a **sieve tube** element, with surrounding **companion cells**.
- Sieve plates contain holes through which phloem sap containing sugars pass through.
- Phloem is surrounded by companion cells as mature phloem cells do not have a nucleus. The phloem cells survive by getting ATP from companion cells through the plasmodesmata.

▼ Outline the importance of water and inorganic ions to plants

Importance of Water:

1. Photosynthesis: Water and CO₂ combine to make glucose and oxygen in the presence of sunlight.
2. Support: Water provides support to non-woody plants via turgor pressure in plant cells. Water moves into cells causing the cytoplasm to be pushed up against the cell membrane - this makes the cell rigid and the plant stay upright.
3. Transport: Mineral ions and sugars are dissolved in water and carried around the plant in the transpiration stream.
4. Acts as a cooling agent - evaporates to cool down the plant.
5. Reactant for chemical reactions such as hydrolysis.

Importance of Mineral Ions:

<u>Aa</u> Name	<u>≡</u> Importance	<u>≡</u> Deficiency
<u>Nitrate</u>	Required in the synthesis of DNA, Proteins, Enzymes, Hormones, etc. They make the nucleotide bases in DNA and amino acids	Stunted Plant growth and yellow leaves
<u>Calcium</u>	Combine with Pectin in the middle lamella to form Calcium Pectate which holds adjacent plant cell walls together.	Death of growing points of plants. Young leaves becoming yellow and crinkly.
<u>=</u>	Determines the permeability of cell membrane	-
<u>Magnesium</u>	Needed to produce chlorophyll → Green pigment that traps light energy and converts solar energy to chem. energy during photosynthesis.	Yellow patches on old leaves, stunted growth
<u>Untitled</u>	Needed to activate enzymes and amino acid synthesis.	-
<u>Phosphate</u>	Required to make ADP and ATP	Dark green-purple leaves and stem. Stunted growth.

▼ Outline the use of plants fibres and starch as a sustainable method to replace oil-based plastics

Sustainability:

Plant material is **renewable and sustainable** - its level can be maintained at a constant level. This is because we can grow plants at the rate of which they are being used, whereas with substances like crude oil, which is formed over millions of years, it is being used at a rate extremely higher than the rate at which it is made, meaning we will soon run out of it. This is why it is important to turn to plant-based substitutes for the current uses of crude oil and fossil fuels, such as the development of **biodiesel** as a substitute for diesel. Biodiesel is made from ethanol synthesised from the fermentation of plants.

Furthermore, plants have a lot of useful qualities such as **tensile strength** provided by sclerenchyma fibres. In the future we could see plant based products replacing oil-based plastics, not only is this advantageous as oil is running out, but also because plant based products are much more **biodegradable** so would reduce the waste problems created by non-biodegradable plastics.

Plants fibres have other uses to humans, for instance:

- Plant fibres are very strong therefore they can be used to make materials, such as **ropes and fabrics**.
- Production of plant fibre products is **cheaper** than oil-based ones
- Starch can be used to make **bioplastics and bioethanol**
- Substances derived from plants, such as garlic and cinnamon sticks, have **antimicrobial and therapeutic properties**



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